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102
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Jharkhand University of Technology, Ranchi

Diploma 1st Semester Examination, 2025 (NEP)

Subject : Engineering Physics | **Subject Code : BSC 102**

Time Allowed : 3 Hours | **Full Marks : 70**

Pass Marks : 21

Answer in your own words.

Answer any five *questions in which* Question No. 1 *is compulsory.*

The figures in the margin indicate full marks.

All questions carry equal marks.

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1. Answer the following multiple-choice questions: 2×7

(i) Which instrument is used to measure very small thickness?

(a) Vernier calipers (b) Screw gauge

(c) Micrometer (d) Compass

(ii) Which of the following is a vector quantity?

(a) Speed (b) Distance

(c) Acceleration (d) Work

(iii) Dimensional formula of force is

(a) $[MLT^{-2}]$ (b) $[ML^2T^{-2}]$

(c) $[MT^{-2}]$ (d) $[LT^2]$

(iv) Which law states that stress is proportional to strain within elastic limit?

(a) Pascal's law (b) Hooke's law

(c) Boyle's law (d) Newton's law

(v) Which mode of heat transfer occurs in solids?

(a) Conduction (b) Convection

(c) Radiation (d) All of these

(vi) The acceleration due to gravity decreases with

(a) Altitude (b) Depth

(c) Latitude (d) All of these

(vii) The frequency of a wave is 500 Hz and wavelength is 0.68 m. Find its velocity.

(a) 340 m/s (b) 300 m/s

(c) 250 m/s (d) 400 m/s

2. (a) Define fundamental and derived quantities with examples.

   (b) Convert 5 m/s velocity into cm/s. 7+7

3. (a) Explain Newton's third law of motion with practical examples.

   (b) A body moving with initial velocity 10m/s accelerates at $3m/s^2$ for 5s. Find final velocity and distance covered. 7+7

4. (a) State Newton's law of gravitation and write its mathematical form.

   (b) Calculate g at a depth of 500 m below Earth's surface (Radius of Earth = 6.37×10^6 m). 7+7

5. (a) Define modulus of rigidity and bulk modulus.

   (b) A rod of length 1m and area $2mm^2$ is stretched by 1mm under a load of 10kg. Calculate Young's modulus. 7+7

6. (a) Explain coefficient of thermal conductivity and its SI unit.

   (b) A rod conducts 200J of heat in 20s. Length = 50cm, area = $1cm^2$, temperature difference = 40°C. Find thermal conductivity. 7+7

7. Write short notes on *any four*: 3·5×4

   (a) Significant figures and types of errors

   (b) Law of conservation of linear momentum

   (c) Atmospheric pressure and gauge pressure

   (d) Modes of heat transmission

   (e) Stationary waves and their characteristics

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$$g_d = g\left(1 - \frac{d}{R}\right)$$

$$= 9.8\left(1 - \frac{500}{6.37 \times 10^6}\right)$$

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Jharkhand University of Technology, Ranchi

Diploma 1st Semester Examination, 2025 (NEP)

Subject : Engineering Chemistry **Subject Code : BSC 103**

Time Allowed : 3 Hours **Full Marks : 70**

Pass Marks : 21

Answer in your own words.
Answer any five *questions in which* Question No. 1 *is compulsory.*
The figures in the margin indicate full marks.
All questions carry equal marks.

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1. Choose the correct answer of the following questions: 2×7=14

(i) The fundamental particle with no charge is

(a) Electron (b) Proton
(c) Neutron (d) Positron

(ii) The rule used to fill electrons in increasing order of energy is

(a) Pauli Principle (b) Hund's Rule
(c) Aufbau Principle (d) Bohr Rule

(iii) The electrolyte that completely dissociates in aqueous solution is

(a) Weak electrolyte (b) Strong electrolyte
(c) Non-electrolyte (d) Organic acid

(iv) The process used to extract metal from ore by reduction is

(a) Concentration (b) Refining
(c) Smelting (d) Roasting

(v) Bakelite is prepared by

(a) Addition polymerization (b) Condensation polymerization
(c) Copolymerization (d) Step-growth reaction

(vi) The material used for thermal insulation is

(a) Steel (b) Glass wool
(c) Copper (d) Brass

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(vii) The major pollutant responsible for greenhouse effect is

(a) SO (b) CO
(c) NH (d) None of these

7

2. (a) Define atom. Explain Bohr's model and limitations.

   (b) Explain formation of covalent and ionic compounds with examples. 7+7

3. (a) Define ionisation. Explain Arrhenius theory of ionisation.

   (b) Explain primary and secondary reactions during electrolysis with examples. 7+7

4. (a) What are metals? Explain mechanical properties of metals.

   (b) Describe the extraction process of any two metals. 7+7

5. (a) Explain preparation, properties and uses of alloys.

   (b) Write notes on Duralumin, German Silver and Monel Metal. 7+7

6. (a) Explain polymerization. Distinguish between thermosetting and thermoplastic polymers.

   (b) Explain natural rubber, its vulcanisation and applications. 7+7

7. Write short notes on *any four* of the following: 3·5×4=14

   (a) Electrochemical Series

   (b) Electrolysis of NaCl

   (c) Synthetic Polymers

   (d) Water Pollution — causes and control

   (e) Thermal Insulating Materials

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ALL

Jharkhand University of Technology, Ranchi

Diploma 1st Semester Examination, 2025 (NEP)

Subject : Engineering Mathematics **Subject Code : BSC 101**

Time Allowed : 3 Hours **Full Marks : 70**

Pass Marks : 21

Answer in your own words.

Answer any five *questions in which* Question No. 1 *is compulsory.*

The figures in the margin indicate full marks.

Marks are given in the right margin.

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1. Choose the correct answer in each of the following: 2×7=14

(i) If $2\begin{pmatrix}3 & 4\\ 5 & x\end{pmatrix}+\begin{pmatrix}1 & 2y\\ 0 & 1\end{pmatrix}=\begin{pmatrix}7 & 0\\ 10 & 5\end{pmatrix}$, then values of x and y are

(a) $x=8, y=0$
(b) $x=2, y=-4$
(c) $x=6, y=2$
(d) $x=5, y=3$

(ii) If $\begin{pmatrix}2x+y\\ x-y\end{pmatrix}=\begin{pmatrix}8\\ 4\end{pmatrix}$, then values of x and y are

(a) $x=4, y=0$
(b) $x=8, y=0$
(c) $x=6, y=2$
(d) $x=5, y=3$

(iii) The distance between the points $A(2,-7)$ and $B(6,3)$ is

(a) 4 units
(b) 6 units
(c) 8 units
(d) 10 units

(iv) The distance between the points $A(2,-3)$ and $B(-6,3)$ is

(a) $\sqrt{29}$ units
(b) $-\sqrt{29}$ units
(c) $4\sqrt{29}$ units
(d) $2\sqrt{29}$ units

(v) If the slope of the line joining the points $A(x,2)$ and $B(6,-8)$ is $\frac{-3}{4}$, then x =?

(a) $\frac{-22}{3}$
(b) $\frac{22}{3}$
(c) $\frac{-2}{3}$
(d) 3

(vi) $\sin\left(\frac{19\pi}{3}\right)$ =?

(a) $\frac{\sqrt{3}}{2}$
(b) $-\frac{\sqrt{3}}{2}$
(c) $\frac{1}{2}$
(d) $-\frac{1}{2}$

(vii) $\operatorname{cosec} 840° = ?$

(a) $\frac{2}{\sqrt{3}}$ (b) $\frac{\sqrt{3}}{2}$

(c) $\frac{1}{2}$ (d) $-\frac{1}{2}$

2. (a) Solve the following system of linear equations (use any method):

$$3x + 4y + 2z = 8$$
$$x - 2y - 3z = 3$$
$$3x - 2y + 6z = -2$$

(b) Find the eigenvalues and eigenvectors of the matrix $A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$. 7+7

3. (a) Find the equation of the line whose portion intercepted between the axes is bisected at the point $(3, -2)$.

(b) Find the equation of the line whose y-intercept is -3 and which is perpendicular to the line joining the points $(-2, 3)$ and $(4, -5)$. 7+7

4. (a) If $\sin\theta = \frac{\sqrt{3}}{2}$ and θ lies in quadrant II, find the values of all the other five trigonometric functions.

(b) Calculate the values of: (i) $\sin 15°$, (ii) $\cos 15°$, (iii) $\tan 15°$. 7+7

5. (a) Differentiate y $w.r.t.$ x:

$$y = \log\left(\operatorname{cosec}\frac{x}{2} + \cot\frac{x}{2}\right)$$

(b) Find the point of local maxima or minima and corresponding local maximum and minimum value for the following function:

$f(x) = -x^3 + 12x^2 - 5$ 7+7

6. (a) Evaluate the integral: $I = \int_0^{\frac{\pi}{2}} \cos^4 x dx$.

(b) Evaluate the integral: $I = \int \left(\frac{x^3+4x^2-3x-2}{x+2}\right) dx$ 7+7

7. Answer *any four* of the following: 3.5×4=14

(a) If $2\begin{pmatrix} x & 5 \\ 7 & y-3 \end{pmatrix} + \begin{pmatrix} 3 & -4 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 7 & 6 \\ 15 & 14 \end{pmatrix}$, then find the value of x and y.

(b) Find the equation of the line perpendicular to the line $x - 7y + 5 = 0$ and having x-intercept 3.

(c) Prove that $\left(\frac{\cos x+\sin x}{\cos x-\sin x} - \frac{\cos x-\sin x}{\cos x+\sin x}\right) = 2\tan 2x$.

(d) A spherical soap bubble is expanding so that its radius is increasing at the rate of 0.02 cm/sec. At what rate is the surface area increasing when its radius is 5 cm? (Take $\pi = 3.14$)

(e) Evaluate the integral: $I = \int \left(\frac{2}{1+x^2} - \frac{2}{\sqrt[3]{x^4}} + 3x^{-1} - \frac{3}{x\sqrt{x^2-1}} + 4e^{\frac{x}{4}} + 5^x\right) dx$

Jharkhand University of Technology, Ranchi

Diploma 1st Semester Examination, 2024 (NEP-2024)

Subject : Engineering Physics Subject Code : BSC 102

Time Allowed : 3 Hours Full Marks : 70

Answer in your own words.
Answer any five *questions.* Question No. 1 *is compulsory.*
Marks are given in the right margin.

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1. Choose the correct answer in the following: 2×7=14

(i) The dimensional formulae for Speed is

(a) $[M^0L^0T^0]$ (b) $[M^0L^1T^{-1}]$

(c) $[M^1L^1T^1]$ (d) None of these

(ii) Stoke's formula is related with

(a) Initial velocity (b) Final velocity

(c) Terminal velocity (d) Critical velocity

(iii) Isobaric process is

(a) No communication of heat (b) Pressure constant

(c) Volume constant (d) Temperature constant

(iv) Laser is

(a) Bichromatic (b) Monochromatic

(c) Polychromatic (d) None of these

(v) Unit of stress is

(a) Newton (b) Joule

(c) N/m^2 (d) None of these

(vi) S.I. unit of temperature is _____

(a) Celsius (b) Kelvin

(c) Newton (d) None of these

(vii) The physical property of a material that can not returns to its original shape after the stress is removed is known as

(a) Plasticity (b) Elasticity

(c) Opacity (d) None of these

2. (a) Define unit. Also explain fundamental unit and derived unit with examples.

   (b) Briefly discuss Vernier calliper with schematic diagram. 7+7

3. (a) Explain Young's modulus of elasticity, Bulk modulus of elasticity and Modulus of rigidity.

   (b) Explain Newton's law of viscosity. 7+7

4. (a) Briefly explain the modes of heat transfer.

   (b) What are Centigrade scale, Fahrenhiet scale and Kelvin scale? 7+7

5. (a) What do you mean by reflection and refraction of light? Explain.

   (b) Derive the relation among velocity, frequency and wavelength. 7+7

6. (a) Explain construction and working of photoelectric cell.

   (b) Calculate the energy of photon of red light in a vacuum with a wavelength of 695nm. 7+7

7. Write short notes on *any four*: 3.5×4=14

   (a) Reynold's number

   (b) Capillary action

   (c) Properties of laser

   (d) X-rays

   (e) Boyle's law

Jharkhand University of Technology, Ranchi

Diploma 1st Semester Examination, 2024 (NEP-2024)

Subject : Engineering Chemistry **Subject Code : BSC103**

Time Allowed : 3 Hours **Full Marks : 70**

Pass Marks : 21

Answer in your own words.
Answer five questions in which Question No. 1 is compulsory.
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1. Choose the correct option: 2×7=14

(i) The electronic configuration of Cu^{2+} ion is

(a) [Ar] $4s^1\ 3d^8$ (b) [Ar] $4s^2\ 3d^{10}\ 4p^1$

(c) [Ar] $4s^1\ 3d^{10}$ (d) [Ar] $3d^9$

(ii) Bohr proposed that while revolving in discrete orbits, the electrons

(a) gain energy (b) lose energy

(c) do not radiate energy (d) first lose energy and then gain energy

(iii) What is the most difficult atom to ionize?

(a) Hydrogen (b) Helium

(c) Beryllium (d) Neon

(iv) Which of the following is a synthetic polymer commonly used in making plastic?

(a) Cellulose (b) Starch

(c) Protein (d) Polyethylene

(v) Which of the following is not a type of electrochemical cell?

(a) Voltaic cell (b) Photovoltaic cell

(c) Electrolytic cell (d) Fuel Cell

(vi) Which industry causes all three: air, water and land pollution?

(a) Fertilizer and pesticides (b) Oil refineries and iron, steel

(c) Oil refineries and caustic soda (d) Iron, steel and caustic soda

(vii) Which of the following metal is utilized in trucks, automobile engines, aircraft and missiles?

(a) Stainless steel (b) Carbon steels

(c) Magnesium (d) Cast irons

2. (a) Write about the different types of valency with suitable examples.

(b) Define orbitals. Explain Aufbau Principle for filling up of the orbitals. 7+7

3. (a) Explain Arrhenius Theory of Ionization.

(b) What is degree of Ionization? Discuss the factors affecting degree of ionization. 7+7

4. (a) What are different types of alloys? Describe the purpose of making alloys.

(b) Explain the physical properties & applications of Cu and Al. 7+7

5. (a) Define polymers. Compare natural and synthetic rubber with suitable examples.

(b) Write about the engineering applications of plastic based on their properties. 7+7

6. (a) What do you mean by E-waste? Write about different types of waste.

(b) Describe air pollution and explain causes and remedial measure of air pollution. 7+7

7. Write short notes on *any four* of the following: 3.5×4=14

(a) Water Pollution

(b) Electrolysis

(c) Gun Metal

(d) Thermal Insulators

(e) BOD and COD

101
ALL

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Subject : Engineering Mathematics **Subject Code : BSC 101**

Time Allowed : 3 Hours **Full Marks : 70**

Answer in your own words.
Answer any five *questions.* Question No. 1 *is compulsory.*
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1. Choose the correct answer in the following: 2×7=14

(i) If A is a 2×2 matrix such that $|A| = 5$ and $|A| \neq 0$ then the value of $|4A|$ is

(a) 20 (b) 25

(c) 80 (d) None of these

(ii) If A and B are invertible square matrices of the same order then $(AB)^{-1} = ?$

(a) AB^{-1} (b) $A^{-1}B^{-1}$

(c) $B^{-1}A$ (d) $B^{-1}A^{-1}$

(iii) Find the slope of a line whose inclination is 60°.

(a) $\sqrt{3}$ (b) $\frac{1}{\sqrt{3}}$

(c) 1 (d) None of these

(iv) The equation of the line that makes intercepts at 2 and –3 on the x-axis and y-axis respectively is represented as

(a) $2x - 3y = 6$ (b) $x - 2y = 3$

(c) $3x - 2y = 6$ (d) None of these

(v) If $\sin x = \frac{1}{6}$ then $\sin 3x$ can be expressed as

(a) $\frac{1}{2}$ (b) $\frac{13}{27}$

(c) $\frac{12}{27}$ (d) None of these

(vi) The first order derivative of $\log_3 x$ is

(a) $\log 3$ (b) $\frac{1}{x}$

(c) $\frac{1}{x(\log 3)}$ (d) None of these

(vii) Find the value of the integral $\int \frac{\sin 2x}{\sin x}\,dx$. (2)

(a) $2\sin x + c$ (b) $2\cos x + c$

(c) $\frac{1}{2}\sin x + c$ (d) $\frac{1}{2}\cos x + c$

2. (a) Prove that $\begin{vmatrix} 1 & b+c & b^2+c^2 \\ 1 & c+a & c^2+a^2 \\ 1 & a+b & a^2+b^2 \end{vmatrix} = (a-b)(b-c)(c-a)$.

(b) Solve the system of equations $x + y + z = 6; 2x + 3y - z = 5; 6x - 2y - 3z = -7$ using Cramer's rule. 7+7

3. (a) Find the equation of the line passing through the point (–2, –4) and perpendicular to the line $3x - y + 5 = 0$.

(b) Reduce the equation $\sqrt{3}x + y + 2 = 0$ to intercept form and find the intercepts on the axes. 7+7

4. (a) Find the values of all trigonometric functions of 120°.

(b) Prove that: $\cos\alpha + \cos\beta + \cos\gamma + \cos(\alpha+\beta+\gamma) = 4\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\beta+\gamma}{2}\right)\cos\left(\frac{\gamma+\delta}{2}\right)$. 7+7

5. (a) Find the second order derivative of $e^{2x}\cos 3x + x^4$.

(b) Obtain the local maxima or local minima of $f(x) = x^3 - 6x^2 + 9x + 15$. Also find the local maximum or local minimum values of $f(x)$. 7+7

6. (a) Evaluate: $\int_{-a}^{a} \sqrt{\frac{a-x}{a+x}}\,dx$.

(b) Calculate the area bounded by the parabola $y^2 = 4ax$ and its latus rectum. 7+7

7. Write short notes on *any four*: 3.5×4=14

(a) Inverse of a matrix

(b) Collinear points

(c) ASTC diagram

(d) Stationary points

(e) Integration by parts